SAGAR SHUKLA

Machine Learning & Computer Vision Engineer | Python, PyTorch, Deep Learning, GenAI

Bhopal, India | +91 9302368472 | sagar1.7shukla@gmail.com | linkedin.com/in/sagar-shukla-3342aa34a | github.com/Saanctity

PROFESSIONAL SUMMARY

B.Tech Computer Science graduate specializing in Machine Learning, Computer Vision, and Generative AI systems. Hands-on experience developing deep learning architectures (PyTorch, U-Net, YOLO, GANs), biometrics/metric embeddings (MTCNN, Inception-ResNet), and production RAG pipelines with cloud deployment on GCP. Proven engineering track record handling complex high-dimensional multi-band tensors, unstructured databases, and full-stack API integration (FastAPI, Flask, Streamlit).

TECHNICAL SKILLS

- **Core Machine Learning & DL:** PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, CNNs, U-Net, YOLO, Transfer Learning, GANs, LSTM, Tokenization
- **Computer Vision & Processing:** OpenCV, Semantic Segmentation, Object Detection, Face Recognition (MTCNN, Facenet), Multi-band Image Tensors
- **GenAI & Agent Systems:** RAG Pipelines, Large Language Models (LLMs), Embeddings, Prompt Engineering, Vector Search, Cloud AI Deployment
- **Languages, APIs & Backend:** Python, SQL, MongoDB, TypeScript, C++, FastAPI, Flask, Streamlit, Dash, RESTful APIs, React
- **Data Engineering & DevOps:** Pandas, NumPy, xarray, Rasterio, Git/GitHub, Docker, GCP (Cloud SQL, Compute Engine), Vercel, Render, Azure AI Foundry, Azure OpenAI Service
- **Evaluation & Metrics:** IoU, Dice Loss, F1-Score, Precision-Recall, Confusion Matrix, Cross-Validation, Latency Optimization

KEY TECHNICAL PROJECTS

FRIDAY AI Assistant — Full-Stack Conversational & System Automation Agent — **github.com/Saanctity/FRIDAY-AI-ASSISTANT**

Technologies: Python, React, GenAI / LLM APIs, Vercel, Render

- Architected and deployed a multi-tier AI assistant with a responsive React frontend hosted on Vercel and an asynchronous Python backend on Render.
- Engineered intent recognition and tool execution pipelines enabling voice/text interaction, real-time web querying, and autonomous local system workflows.
- Implemented structured API error handling, session persistence, and latency-optimized endpoint communication.

Custom U-Net Multi-Band Tensor Segmentation Pipeline — **github.com/Saanctity/unet-water-segmentation-sentinel2**

Technologies: PyTorch, Deep Learning, High-Dimensional Arrays

- Engineered an end-to-end 117K-parameter U-Net architecture from scratch with encoder-decoder skip connections to segment high-dimensional image tensors.
- Built custom PyTorch DataLoaders to ingest multi-band array data; achieved 0.65 IoU and 0.79 F1-score on held-out geographic test sites.
- Documented performance boundaries, metric tracking, and loss convergence using combined Binary Cross-Entropy and Dice loss functions.

Medical Image Synthesis with Generative Adversarial Networks (GANs)
Technologies: PyTorch, Streamlit, Generative AI

- Constructed and trained a Deep Convolutional GAN (DCGAN) to synthesize photorealistic medical imaging data (X-rays, MRIs) to mitigate severe dataset scarcity.
- Demonstrated downstream classifier accuracy improvements when augmented with synthetic generator outputs; built an interactive Streamlit testing web application.

High-Throughput Face Biometrics & Metric Embeddings
Technologies: MTCNN, Inception-ResNet V1, Flask, PyTorch

- Engineered a production face detection and recognition pipeline using MTCNN for bounding-box extraction and Inception-ResNet V1 for 512-D vector embeddings.
- Constructed a vector lookup library evaluating cosine distance thresholds and wrapped the complete inference engine in a scalable Flask web API.

Real-Time Video Object Detection & Tracking
Technologies: YOLO, OpenCV, Video Processing

- Trained and fine-tuned a custom YOLO architecture on dense video feeds (Dhaka traffic dataset), parsing XML Pascal-VOC annotations.
- Implemented robust frame extraction, bounding-box coordinate transformations, and data augmentations to generalize under severe occlusion.

PROFESSIONAL EXPERIENCE

Symbioverse — Volunteer Data & Web Analyst **Nov 2025 – Apr 2026**

- Deployed and maintained a production RAG-based AI research engine on Google Cloud Platform (Cloud SQL, Compute Engine) across custom domain setups.
- Diagnosed and resolved authentication routing bugs across public application endpoints; engineered data pipelines to structure scientific variables.

IISPR — Deep Learning & NLP Research Intern **Jul 2025 – Oct 2025**

- Developed multi-model sentiment architectures (VADER, LSTM, CNN) on real-world datasets and prototyped automated conversational systems.

Unified Mentor — Machine Learning Intern **Jun 2025 – Sep 2025**

- Engineered and evaluated 10 end-to-end ML/DL models covering medical imaging diagnostics (lung, liver, thyroid), fraud detection, and multi-class image classifiers.

EDUCATION & CERTIFICATIONS

- **B.Tech in Computer Science Engineering:** Radharaman University, Bhopal | 2022 – 2026
- **Applied Data Science & Computer Vision Labs:** WorldQuant University | GANs, Inception-ResNet, YOLO, Time-Series
- **Remote Sensing & SAR Applications:** NASA ARSET (Fundamentals & Surface Water Monitoring), 2026
- **Languages:** English (Fluent), Hindi (Fluent), Japanese (N4 Level), Spanish (B1)